EXPERIMENT 11 : Simple Top-Down Parser for Context-Free Grammar

Aim: To implement a simple top-down parser for a Context-Free Grammar (CFG) using Python.

Program:

```
grammar = {
    "S": [["NP", "VP"]],
    "NP": [["Det", "N"]],
    "VP": [["V", "NP"]],
    "Det": [["the"], ["a"]],
    "N": [["student"], ["teacher"], ["book"]],
    "V": [["reads"], ["likes"]]
}
def parse(symbol, words, index):
    if symbol not in grammar:
        if index < len(words) and symbol == words[index]:
            return index + 1
        return None
    for rule in grammar[symbol]:
        current = index
        success = True
        for item in rule:
            current = parse(item, words, current)
            if current is None:
                success = False
                break
        if success:
            return current
    return None
```

```
sentence = input("Enter a sentence: ").lower()
words = sentence.split()
result = parse("S", words, 0)
if result == len(words):
    print("Sentence is accepted by the grammar.")
else:
    print("Sentence is rejected by the grammar.")
```

Output: